

(1.1) Dokažte, že v libovolném komplexním vektorovém prostoru V se skalárním součinem $\langle, \rangle$ platí (pro libovolné $u, v \in V$)

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2),$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4}(\|u - iv\|^2 - \|u + iv\|^2),$$

kde $\operatorname{Re}(x)$ značí reálnou a $\operatorname{Im}(x)$ imaginární část komplexního čísla x . (Pro inspiraci vizte tvrzení 8.32 ve skriptech.)

$$\begin{aligned} \|u + v\|^2 &= \langle u + v, u + v \rangle = \\ &= \langle u + v, u \rangle + \langle u + v, v \rangle = \overline{\langle u, u + v \rangle} + \overline{\langle v, u + v \rangle} = \\ &= \overline{\langle u, u \rangle} + \overline{\langle u, v \rangle} + \overline{\langle v, u \rangle} + \overline{\langle v, v \rangle} = \|u\|^2 + \overline{\langle u, v \rangle} + \langle u, v \rangle + \|v\|^2 = \\ &= \|u\|^2 + 2\operatorname{Re}(\langle u, v \rangle) + \|v\|^2 \end{aligned}$$

definice $\| \cdot \|$
definice $\langle \cdot, \cdot \rangle$

$$z + \bar{z} = 2\operatorname{Re}(z)$$

$$\text{dk: } a + bi + a - bi = 2a$$

$$\begin{aligned} \|u - v\|^2 &= \langle u - v, u - v \rangle = \\ &= \langle u - v, u \rangle + \langle u - v, -v \rangle = \langle u - v, u \rangle - \langle u - v, v \rangle = \\ &= \overline{\langle u, u - v \rangle} - \overline{\langle v, u - v \rangle} = \overline{\langle u, u \rangle} - \overline{\langle u, v \rangle} - \overline{\langle v, u \rangle} + \overline{\langle v, v \rangle} = \\ &= \|u\|^2 - (\overline{\langle u, v \rangle} + \langle u, v \rangle) + \|v\|^2 = \|u\|^2 - 2\operatorname{Re}(\langle u, v \rangle) + \|v\|^2 \end{aligned}$$

$$\begin{aligned} \|u + v\|^2 - \|u - v\|^2 &= \|u\|^2 + 2\operatorname{Re}(\langle u, v \rangle) + \|v\|^2 - (\|u\|^2 - 2\operatorname{Re}(\langle u, v \rangle) + \|v\|^2) = \\ &= 4\operatorname{Re}(\langle u, v \rangle) \end{aligned}$$

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2)$$

$$\begin{aligned} &= \langle u - iv, u - iv \rangle = \langle u - iv, u \rangle + \langle u - iv, -iv \rangle = \\ &= \langle u - iv, u \rangle - i\langle u - iv, v \rangle = \overline{\langle u, u - iv \rangle} - i\overline{\langle v, u - iv \rangle} = \\ &= \overline{\langle u, u \rangle} - i\overline{\langle u, v \rangle} - i(\overline{\langle v, u \rangle} - i\overline{\langle v, v \rangle}) = \\ &= \|u\|^2 + i\overline{\langle u, v \rangle} - i\langle u, v \rangle - i^2\|v\|^2 = \\ &= \|u\|^2 + i(\overline{\langle u, v \rangle} - \langle u, v \rangle) + \|v\|^2 = \\ &= \|u\|^2 + i(-2i\operatorname{Im}(\langle u, v \rangle)) + \|v\|^2 = \\ &= \|u\|^2 + 2\operatorname{Im}(\langle u, v \rangle) + \|v\|^2 \end{aligned}$$

def $\| \cdot \|$
def $\langle \cdot, \cdot \rangle$

$$z - \bar{z} = 2i\operatorname{Im}(z); \quad \bar{z} - z = -2i\operatorname{Im}(z)$$

$$\text{dk: } a + bi - (a - bi) = 2bi$$

$$\begin{aligned} \|u + iv\|^2 &= \langle u + iv, u + iv \rangle = \langle u + iv, u \rangle + \langle u + iv, iv \rangle = \\ &= \langle u + iv, u \rangle + i\langle u + iv, v \rangle = \overline{\langle u, u + iv \rangle} + i\overline{\langle v, u + iv \rangle} = \\ &= \overline{\langle u, u \rangle} + i\overline{\langle u, v \rangle} + i(\overline{\langle v, u \rangle} + i\overline{\langle v, v \rangle}) = \\ &= \|u\|^2 + i\overline{\langle u, v \rangle} + i\langle u, v \rangle - \|v\|^2 = \\ &= \|u\|^2 + i(\overline{\langle u, v \rangle} + \langle u, v \rangle) + \|v\|^2 = \\ &= \|u\|^2 + 2\operatorname{Re}(\langle u, v \rangle) + \|v\|^2 \end{aligned}$$

$$\begin{aligned}\|u - v\|^2 - \|u + v\|^2 &= \|u\|^2 + 2\operatorname{Re}\langle u, v \rangle + \|v\|^2 - (\|u\|^2 - 2\operatorname{Re}\langle u, v \rangle + \|v\|^2) = \\ &= 4\operatorname{Re}\langle u, v \rangle\end{aligned}$$

$$\operatorname{Re}\langle u, v \rangle = \frac{1}{4}(\|u - v\|^2 - \|u + v\|^2)$$